

### LESSON 113 (1.5 GROUND)

#### Lesson Objective

During this lesson, the student will become familiar with the regulations applicable to a student pilot preparing for solo flight including both 14 CFR part 61 and 91. This will include a detailed discussion on student pilot privileges and limitations, endorsements required for solo flight, airworthiness requirements and required certificates and documents. At the end of the lesson, complete the FAA student pilot application and guide the student on how to obtain a medical.

#### Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
  - Quiz the student on the relevant sections of the Flight Operations Manual:
    - Part 02, Section 02 (Weather minimums for student pilot).
    - Part 03, Section 03 (Student pilot solo).
  - Quiz the student on the readings as needed.
    - Review the Pilot's Handbook of Aeronautical Knowledge:
      - Chapter 15 - Airspace.
- Student questions.

#### References

- Advisory Circular. ([Search Tool](#))
  - Parts 91 and 135 Single Pilot, Flight School Procedures During Taxi Operations. ([AC 91-73](#))
  - Certification of Pilots and Flight and Ground Instructors. ([AC 61-65](#))
- Federal Aviation Regulations.
- Aeronautical Information Manual ([AIM](#)).
  - Chapter 07 - Safety of Flight.
    - Section 04 - Wake Turbulence.
    - Section 06 - Potential Flight Hazards.
- FIT Flight Operations Manual.
  - Part 02 - Technical.
    - Section 02 - Ramp Operations.
    - Section 03 - Flight Operations.
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
  - Chapter 05 - Aerodynamics of Flight.
  - Chapter 09 - Flight Manuals and Other Documents.
  - Chapter 15 - Airspace.
- [https://www.faa.gov/airports/runway\\_safety/resources/runway\\_incursions/](https://www.faa.gov/airports/runway_safety/resources/runway_incursions/)

### Lesson Content

#### Regulations Applicable to Student Pilots:

- Ensure that the student has the FARAIM opened and follows along.
- Before going through the regulations, highlight and explain the different parts and subparts of the FAR.
  - Part 1: definitions and abbreviations.
  - Part 23: airworthiness standards for normal category airplanes.
  - Part 43: Maintenance, preventive maintenance, rebuilding, and alteration.
  - Part 61: Certification: Pilots, flight instructors, and ground instructors.
    - Subpart A: General
    - Subpart B: Ratings
    - Subpart C: Student Pilot
    - Subpart D: Recreational Pilot
    - Subpart E: Private Pilot
    - Subpart F: Commercial Pilot
    - Subpart G: ATP
    - Subpart H: CFI
  - Part 67: Medical standards and certification.
  - Part 71: Designation of class A, B, C, D, and E airspace areas; air traffic service routes; and reporting points.
  - Part 73: Special use airspace.
  - Part 91: General operating and flight rules.
  - Part 141: Pilot schools.
    - Differentiate between part 61 and part 141.
- Part 61
  - Required documents - [61.3](#):
    - Pilot certificate, medical, and government photo identification when operating as a required pilot.
  - Drugs and alcohol - [61.15](#) and [61.16](#):
    - Outline the consequences.
  - Student pilot certificate duration - [61.19](#):
    - No expiration date if issued after April 1st 2016.
  - Medical certificate classes and durations - 61.23:
    - Special issuance, SODA, etc. (67.401).
  - Type rating of aircraft [61.31](#)
    - Aircraft need a type rating if:
      - Large Aircraft (more than 12,500 lbs)
      - Turbojet-Powered Aircraft
      - Any aircraft deemed necessary by the administrator
  - Pilot Logbooks - [61.51](#):

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- Information to be logged.
- Medical Deficiency - [61.53](#):
  - What constitutes a medical deficiency.
- Flight Review - [61.56](#):
  - Every 24 calendar months.
  - At least an hour ground and an hour flight that must cover the legally required areas.
  - Must receive endorsement to complete.
- Recent flight experience - [61.57](#):
  - Acting as PIC while carrying passengers during day time vs night time.
  - Same category, class, and type.
  - If tailwheel, needs to be to a full stop.
  - At night to a full stop.
- Change of address - [61.60](#):
  - The FAA must be notified in writing within 30 days.
- Solo requirements - [61.87](#):
  - This will be covered under endorsements required for solo flight below.
- General limitations (SP) - [61.89](#):
  - This will be covered under student pilot limitations FAA and FIT below.
- Solo operations in class b airspace - [61.95](#):
  - This will be covered under endorsements required for solo flight below.
- Part 91
  - Pilot in Command (PIC) responsibilities - [91.3](#):
    - PIC is responsible for operation of the aircraft, safety of flight, and has the final authority.
    - This includes determining airworthiness.
  - Aircraft airworthiness - [91.7](#):
    - Cannot fly unless airworthy.
  - Flight manual, placards, and markings - [91.9](#):
    - All must be on the airplane to be airworthy.
  - Careless and reckless operation - [91.13](#):
    - May not operate in a manner that endangers others, on the ground or in the air.
  - Dropping objects - [91.15](#):
    - Can drop objects if the right precautions are taken.
  - Drugs and alcohol - [91.17](#):
    - Cannot be under the influence.
    - 8 hours “bottle to throttle” and a maximum BAC of 0.04%.
    - Compared to FIT’s regulation.
    - Must submit to a drug/alcohol test if requested.
    - Must furnish results to the FAA within 24 hours.
  - Preflight action - [91.103](#):

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- IFR flight or away from the vicinity of an airport.
  - NOTAMs.
  - Weather.
  - Known ATC delays.
  - Runway lengths.
  - Alternates if needed.
  - Fuel requirements.
  - Takeoff/landing data.
- Crewmembers at stations
  - [91.105](#):
  - When must required crew members be at their stations?
  - When should seatbelts and shoulder harnesses be worn?
- Use of safety belts and harnesses
  - [91.107](#):
  - All occupants must be briefed on how to use seatbelts and shoulder harnesses.
  - When should seatbelts and shoulder harnesses be worn?
- Operating near other aircraft
  - [91.111](#):
  - May not operate an aircraft so close to another that it creates a collision hazard.
  - Review formation flight.
- Right of way
  - [91.113](#):
  - See and avoid other aircraft if the weather permits!
  - Review all of the different scenarios and who has the right of way.
- Speed limits
  - [91.117](#):
  - 250 kts below 10,000 ft MSL (Applies in Bravo).
  - 200 kts below Class B or in VFR Corridor.
  - 200 kts within 4 nm and 2,500' Class D or C primary airport.
  - May use minimum safe speed if needed.
- Minimum safe altitudes
  - [91.119](#)
  - Congested
    - 1,000 ft above, 2,000 ft horizontally.
  - Other than congested
    - 500 ft above.
  - Sparsely populated or open water
    - cannot operate within 500' of the nearest vessel, structure, person, or vehicle.
- Altimeter Settings
  - [91.121](#)

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- Compliance with ATC
  - [91.123](#):
  - Must obey instructions unless amended clearance or emergency.
  - Must notify ATC immediately when deviating.
  - May need to submit a report to ATC within 48 hours.
- Temporary Flight Restrictions (TFRs) - 91.137-145:
  - These are where TFR NOTAMs derive authority.
- FAA fuel reserves
  - [91.151](#):
  - You must have fuel to fly from departure to destination, destination to any alternates, and then fly thereafter at cruise power setting for:
    - Day: 30 mins.
    - Night: 45 mins.
    - FIT: 60 mins day or night.
- VFR cruising altitudes
  - [91.159](#):
  - Purpose is for separation on converging courses.
  - Based off of the magnetic course starting at or above 3,000 ft
    - 360-179 degrees → odd 1000's + 500.
    - 180 - 359 degrees → even 1000's + 500.
- Required aircraft equipment
  - [91.205](#)
  - This will be further explained in the airworthiness section.
- Emergency Locator Transmitters
  - [91.207](#)
  - When are they needed?
  - How often must they be inspected?
  - What are they inspected for?
  - How often do you need to change the battery?
- Aircraft lights
  - [91.209](#):
  - Used while the aircraft is being operated.
  - May be turned off when they are unsafe.
- Supplemental oxygen
  - [91.211](#)
  - For unpressurized aircraft
    - From 12,500 ft MSL to 14,000 ft MSL
      - The minimum flight crew is provided with and uses supplemental oxygen for the part of the flight at those altitudes that is more than 30 minutes of duration.
    - Above 14,000 ft MSL to 15,000 ft MSL

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- The minimum flight crew uses supplemental oxygen.
- Above 15,000 ft MSL
  - The minimum flight crew uses supplemental oxygen and each occupant is provided with supplemental oxygen.
- Inoperative equipment
  - [91.213](#)
  - This will be further explained in the airworthiness section.

### Student Pilot Limitations FAA and FIT

- Review FAA student pilot limitations in [61.89](#).
- Review student pilot limitations in the Flight Operations Manual (Version 6):
  - Part 02, Section 02 (Weather minimums for student pilot).
  - Part 03, Section 03 (Student pilot solo).

### Personal Minimums

- PAVE checklist
  - Pilot.
  - Airplane.
  - enVironment.
  - External pressures.
- IMSAFE checklist
  - Illness.
  - Medication.
  - Stress.
  - Alcohol.
  - Fatigue.
  - Emotions/eating.
- The student should develop their own minimums depending on their comfort level.

### Go/No-Go Decision

- The student must make a go/no-go decision depending on:
  - Personal minimums.
  - FIT minimums.
  - Aircraft.
  - Airspace.
- Review the students planning with a focus on:
  - Weather, weight and balance, performance, flight planning, and fuel calculations.
  - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
  - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
  - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is

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continuous and should be a consideration even once airborne.

- At this stage of flight training, the student should be able to list all of the factors affecting a go/no-go decision.

### Endorsements Required for Solo Flight

- Required pre-solo endorsements - [61.87](#).
  - Pre-solo aeronautical knowledge test - 61.87(b).
  - Pre-solo flight training - 61.87(c).
  - Solo flight (initial 90 days) - 61.87(n).
- Require solo cross-country endorsements - 61.93.
  - Review the solo cross-country general requirements.
  - Cross-country flight training - 61.93(c)1(c)2.
  - Solo cross-country flight (specific route) - 61.93(c)3.
- Additional endorsements - 61.87, 61.93, and 61.95.
  - Solo flight (additional 90 days) - 61.87 (p).
  - Solo takeoffs and landings at another airport within 25NM - 61.93 (b)(1).
  - Repeated Solo cross-country flights not more than 50 NM from the departure airport - 61.93(b)(2).
  - Solo flight in class B airspace - 61.95(a).
  - Solo flight to, from or at an airport in class B airspace - 61.95(a). and 61.95(b).
  - Pre-solo flight training at night - 61.87(c) and (o).

### Airworthiness Requirements

- Define airworthiness.
  - Type Certificate Data Sheet.
    - Ensure the student knows where to find it.
      - Faa.gov/aircraft → technical information → TCDS → search your aircraft.
  - Supplemental Type Certificate.
  - Inspections (AVIATES):
    - Annual.
    - VOR.
    - 100 hours.
    - Altimeter.
    - Transponder.
    - ELT.
    - Static system.
  - Airworthiness directives:
    - Define what an airworthiness directive is.
    - Three types:
      - Notice of proposed rulemaking.
      - Final rule.

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- Emergency.
- Service bulletins:
  - Define what a service bulletin is and how it is different from an AD.
- Required documents and equipment for VFR flight.
  - Applicable regulations: 91.203 and 91.205.
  - Required documents (ARROWES):
    - Airworthiness certificate.
      - Does not expire but invalid if not airworthy.
      - Check tail number and signature during pre-flight.
    - Registration certificate.
      - Typically expires every 7 years.
      - Check tail number and expiration date during pre-flight.
      - Different ways to lose the registration certificate:
        - Death.
        - Transfer.
        - Aircraft is found doing illegal activities.
        - Aircraft is totaled.
        - Certificate holder requests in writing to cancel their registration.
        - Certificate holder loses US Citizenship.
    - Radio station license / radio operator permit.
      - Obtained through the Federal Communications Commission (FCC).
      - Radio station license is for the airplane.
        - Expires every 10 years.
      - Radio operator permit is for the pilot.
        - Does not expire.
    - Operational limitations.
      - POH section 02.
    - Weight and balance.
      - POH section 06.
    - Equipment list.
      - POH section 02.
      - Difference between MEL and KOEL.
    - Supplements (FLICES):
      - FAA form 337.
        - Major alterations.
        - Examples of major alterations can be found in part 43.
      - Life-limited parts.
        - Any part for which a mandatory replacement limit is specified in the type design or the maintenance manual.
      - Inoperative equipment.
        - Refer to the inoperative equipment section below.



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- Compass deviation card.
- External data plate.
  - Located on the left side of the fuselage by the tail.
- Supplemental type certificate.
- Required equipment:
  - Day VFR required instruments (ATOMATOFLAMES) - 91.205.
    - Airspeed Indicator.
    - Tachometer.
      - For each engine.
    - Oil Pressure.
    - Manifold Pressure.
      - For each altitude engine (turbocharged/supercharged).
    - Altimeter.
    - Temperature Gauge.
      - For each liquid cooled engine.
    - Oil temperature.
    - Fuel Quantity Gauge.
    - Landing Gear Position Lights.
      - If retractable.
    - Anti-Collision Lights.
      - Aircraft certified after March 11<sup>th</sup>, 1996.
    - Magnetic direction indicator.
    - ELT.
    - Seatbelts.
  - Night VFR required instruments (FLAPS) - 91.205.
    - Fuses/resettable circuit breakers.
    - Landing Light (Electric) for hire.
    - Anti-Collision Lights.
      - Aircraft certified after August 11<sup>th</sup>, 1971.
      - Aircraft certified before that date STILL NEED anti-collision lights, but the standards/requirements are different.
    - Position Lights.
    - Source of Electricity (Alternator/Battery).
- Inoperative equipment - 91.213:
  - If an aircraft has a Minimum Equipment List, then refer to that first and find out if the aircraft is still airworthy.
  - Determining Airworthiness with inoperative equipment without MEL (TARK):
    - Type certificate data sheet (TCDS).
    - Airworthiness Directives.
    - Required for 91.205.

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- KOEL, Kinds of Operating Equipment List (section 2 of POH).
- If the instrument or equipment is NOT required:
  - Remove from the airplane in accordance to 43.9 or deactivate and placard inoperative.

### Certificates and Documents

- Medical certificates - 61.23.
  - This was reviewed under part 61 regulations above.
  - Review BasicMed.
- Student pilot certification - 61.83.
  - Eligibility requirements - 61.83.
  - Application - 61.85.
  - Required endorsements were reviewed above.
- Private pilot certification - 61.102.
  - Eligibility requirements - 61.103.
  - Aeronautical experience - 61.109
    - Highlight the differences between part 61 and part 141.
  - Endorsements required for check ride/end-of-course.
    - Aeronautical knowledge test - 61.35(a), 61.103(d), and 61.105.
    - Flight proficiency - 61.103(f), 61.107(b), and 61.109.
    - Prerequisite for practical test - 61.39(a)(6)(iii).
  - Privileges and limitations - 61.113.
- Require logbook entries - 61.51.

### Airspace (BASIC INTRODUCTION – THE REST WILL BE INTRODUCED IN STAGE 2)

- Controlled airspace (91.127 - 91.135).
  - General term that refers to the different classifications of airspace within which air traffic control (ATC) service is provided. (STAGE 1 WILL ONLY FOCUS ON THE FOLLOWING):
    - Class D:
      - Generally, from SFC to 2,500 ft MSL.
      - Surrounds airports that have a control tower.
      - Configuration can be different depending on the airport (KMLB!).
      - Usually designed to contain instrument approaches.
      - Entry requirements:
        - Establish two-way radio communication (ATC MUST say your call-sign).
      - Required equipment:
        - Two-way radios.
    - Class E:
      - Controlled airspace that is not class A, B, C, or D.

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- Extends up from either the surface or a designated altitude to the overlying or adjacent controlled airspace.
- Sectional and other charts depict all locations of Class E airspace with bases below 14,500 feet MSL. In areas where charts do not depict a class E base, class E begins at 14,500 feet MSL.
- In most areas, the Class E airspace base is 1,200 feet AGL. In many other areas, the Class E airspace base is either the surface or 700 feet AGL.
- All airspace above FL 600 is Class E airspace.
- Required equipment:
  - ADSB-OUT: at or above 10,000 feet msl, excluding airspace at and below 2,500 feet agl; over the Gulf of Mexico, at and above 3,000 feet msl, within 12 nm of the U.S. coast.
  - Mode C transponder: at and above 10,000 feet msl, excluding the airspace at and below 2,500 ft agl.
- Uncontrolled airspace (91.126)
  - Portion of the airspace system that is not designated as A, B, C, D, or E.
    - Class G:
      - Uncontrolled.
      - ATC has no authority or responsibility.
        - Although ATC has no authority, a pilot cannot be careless or reckless as this would break the 91.13 regulation.
      - VFR rules apply.
- VFR weather minimums:

| Basic VFR Weather Minimums |                                                                         |                                                |                                                                   |
|----------------------------|-------------------------------------------------------------------------|------------------------------------------------|-------------------------------------------------------------------|
| Airspace                   |                                                                         | Flight Visibility                              | Distance from Clouds                                              |
| Class A                    |                                                                         | Not applicable                                 | Not applicable                                                    |
| Class B                    |                                                                         | 3 statute miles                                | Clear of clouds                                                   |
| Class C                    |                                                                         | 3 statute miles                                | 1,000 feet above<br>500 feet below<br>2,000 feet horizontal       |
| Class D                    |                                                                         | 3 statute miles                                | 1,000 feet above<br>500 feet below<br>2,000 feet horizontal       |
| Class E                    | At or above 10,000 feet MSL                                             | 5 statute miles                                | 1,000 feet above<br>1,000 feet below<br>1 statute mile horizontal |
|                            | Less than 10,000 feet MSL                                               | 3 statute miles                                | 1,000 feet above<br>500 feet below<br>2,000 feet horizontal       |
| Class G                    | 1,200 feet or less above the surface (regardless of MSL altitude).      | Day, except as provided in section 91.155(b)   | 1 statute mile                                                    |
|                            |                                                                         | Night, except as provided in section 91.155(b) | 3 statute miles                                                   |
|                            | More than 1,200 feet above the surface but less than 10,000 feet MSL.   | Day                                            | 1 statute mile                                                    |
|                            |                                                                         | Night                                          | 3 statute miles                                                   |
|                            | More than 1,200 feet above the surface and at or above 10,000 feet MSL. | Day                                            | 5 statute miles                                                   |
|                            |                                                                         | Night                                          | 5 statute miles                                                   |

Figure 15-8. Visual flight rule weather minimums.

- Special use airspace:
  - Where certain activities must be confined.
  - Limitations may be imposed on operations not participating.

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- Information for airspaces is located on aeronautical charts.
- ONLY FOCUS ON THE FOLLOWING FOR STAGE 1:
  - Restricted area:
    - Area in which operations are hazardous to non-participating aircraft.
      - Artillery firing, guided missiles.
    - Flight of aircraft may be restricted, but not prohibited.
    - Must have permission from ATC or the controlling agency to enter.
    - Charted with an “R” followed by a number.
    - Information found on the back of charts.
  - Military Operation Area (MOA):
    - Defined vertical and lateral limits.
    - Separates certain military training activities from IFR traffic.
    - IFR traffic may be cleared through airspace if ATC can provide separation.
    - Additional information for each found on the back of the sectional.
- Other airspace (ONLY FOCUS ON THE FOLLOWING):
  - Parachute jump aircraft operations:
    - Airports at which there are high numbers of parachute operations.
    - Published in chart supplement under Parachute Jumping Areas and depicted on sectional charts.
  - Temporary flight restrictions (TFRs):
    - Designated by flight data center (FDC) Notice to Airmen (NOTAM):
      - Begins with phrase “FLIGHT RESTRICTIONS” followed by location, effective time period, and defined in statute miles, and altitudes affected.
    - Purposes may include:
      - Protect persons or property in air or on surface from existing or imminent hazard.
      - Provide a safe environment for operation of disaster relief aircraft.
      - Prevent unsafe congestion of sightseeing aircraft above incident or event.
      - Protect declared national disasters for humanitarian reasons in Hawaii.
      - Protect the President, Vice President, or other public figures.
      - Provide a safe environment for space agency operations.
    - If penetrated, the pilot may undergo security investigations and certificate suspensions.

### Wake Turbulence Avoidance

- Explain how wake turbulence is created:

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- Pressure differential → low pressure above and high pressure below the wing.
- Air moves from the high-pressure area under the wing to the low-pressure area over the wing.
  - It goes around the wing tip and then trails behind.
  - As air curls around the tip, it combines with the wash to form a fast spinning trailing vortex.
  - Right wing tip - counterclockwise.
  - Left wing tip - clockwise.
- Explain how to avoid wake turbulence:
  - Vortices are greatest when an aircraft is in a heavy, slow, and clean configuration.
    - Avoid flying through another aircraft's flightpath.
    - Rotate prior to proceeding aircraft.
    - Delay takeoff by 2-3 minutes.
    - Avoid following aircraft on similar flight paths.
    - Land past the touchdown point of the other aircraft.
  - Wind is an important factor to wake turbulence.

### Runway Incursion Avoidance

- Define a runway incursion.
- Highlight the 4 categories (INTRODUCE):
  - Category A: serious incident in which a collision was narrowly avoided.
  - Category B: incident in which separation decreases and there is a significant potential for collision.
  - Category C: incident characterized by ample time and/or distance to avoid a collision.
  - Category D: incident that meets the definition of runway incursion such as incorrect presence of a single vehicle/person/aircraft on the protected area of a surface designated for the landing and takeoff of aircraft but with no immediate safety consequences.
- Explain how to avoid a runway incursion:
  - Always brief taxi instructions.
  - Be familiar with the airport layout.
  - Know where hold-short lines are located.
  - Do not cross a hold-short line without obtaining clearance.
  - Ensure the runway is clear of vehicle/person/aircraft before taking-off and landing.

### Visual scanning/Collision avoidance

- Quiz the student on how to scan and avoid traffic.
- Quiz the student on right of way rules.
- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!

### FAA Student Pilot Application

- Complete the student pilot application on IACRA with the student.

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- Once the 8710 is complete, have it looked over by a check airman or an ACI PRIOR to submitting the application.
- Once submitted, inform the student to periodically check (every 2 days should be fine) their IACRA to see if the student pilot certificate has been generated. It usually takes about two weeks for the temporary student pilot certificate to be processed.
- Once the student sees their temporary student pilot certificate has been processed, they can print it from IACRA.
  - They must keep take it to PAS so that a copy can be made.
  - Ensure that the student keeps it on them while flying.
  - The temporary certificate is ONLY VALID FOR 120 DAYS!

### FAA Medical

- Guide the student on how to apply for an FAA medical certificate if they do not already have one.
- They need to fill out a MedExpress form (on their own time).
- Once that is done, they must print it.
- Once that is done, they need to book an appointment with an AME (a list can be found on the FAA website).
- The MedExpress form must be brought to the AME appointment.

### Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Schedule the next lesson.
- Preview the next lesson → 114.
- Assign homework:
  - Tab out the FAR/AIM with all of the regulations spoken about in this lesson.
  - Study! The next ground is the mock!

### Completion Standards

This lesson is complete when the student is able to:

1. Explain regulations applicable to student pilots.
2. List the endorsements required for solo flight and where they should be located.
3. Understand airworthiness requirements and how to determine if a flight can be completed with inoperative equipment.
4. Explain each document and certificate which must be present for both the pilot and aircraft.
5. Explain wake turbulence, its location, effects and avoidance procedures.
6. Describe a runway incursion and mitigation techniques.
7. Describe proper techniques for visual scanning and its importance during all phases of flight.